

## **Gary Chang**

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## **EXPERIENCE**

### **Research / Teaching Assistant; University of Texas at Dallas; 9/2020 – Present; Dallas**

- Accomplished and shortlisted as in top 7 in American Airlines' graduate analytics challenge with an end-to-end misconnection risk pipeline processing 7M+ connection pairs over 10 years of BTS data, applying logit regression, Bayesian hierarchical modeling (INLA), XGBoost, and OR-Tools integer programming to achieve 98% portfolio risk reduction at DFW hub
- Researched factional shifts in Chinese Communist Party leadership using logit, Cox survival, Bayesian/INLA, and network analysis, with LLM/GPT API for content disambiguation and redundancy reduction
- Analyzed spatio-temporal mortality patterns of renal disease in Texas using US Census, NHGIS, and USRDS data with GLS, GLM, GLMM, and Bayesian inference (INLA)
- Detected pre-fabricated tubes in drone imagery using Label-Studio and Roboflow for dataset labeling, YOLO model training, and confidence-threshold-based evaluation
- Estimated typhoon damage at Wenchang Space Center using Sentinel-2 and Google Earth Engine imagery, applying PCA-based atmospheric correction, NDVI/NDHI spectral indices, and radiometric enhancement
- Conducted multi-criteria geographic suitability analysis for co-location of solar and wind energy sites in Tanzania, integrating spatial, socioeconomic, and environmental datasets

### **Senior Software Engineer; WheelsAmerica; 8/2018 – 5/2021; Dallas**

Designed and developed Codegen for data mappings across Ebay, Amazon, SureDone, and company's inventory and accounting systems, developed RESTful interfaces by using PHP, Falcon, AWS, Wordpress, Google Map, and Cloudinary.

### **Marketing Advisor; VSO/UN; 3/2016 – 07/2018; Mwanza, Tanzania**

Worked in T-LED program, Tanzania Local Enterprise Development, to help local SMEs grow their businesses in Malawi, Rwanda, Uganda, and Tanzania by running surveys and market research, delivering training classes, developing business and marketing plans, fund-raising proposals, brochures and posters, and websites.

### **Foreign Scholar; Peking University; 9/2013 – 2/2016; Beijing, China**

- Gave lectures to graduate students at the School of Software and Microelectronics on Professionalism and General Corporate Skills, from Debate, Panel Discussion, Proposal Writing, Reporting, Interview, Strategic Marketing Planning, and Presentation;
- Led the "Classroom Automation" project, by using Linux/Apache/Mysql/Php, which included mobile submission, auto grading, resource sharing, and student social community for the "National Challenge Cup Tournament".

### **Senior Director of Globalization; American Express; 2/2010 – 5/2013; New York**

Worked with various stakeholders and overseas partners to ensure the accomplishment of

deployments of online payment solution, SERVE, by meeting regulatory requirements, business standards, and technical specifications in various areas, including legal, policy compliance, I18N, L10N, IT infrastructure, security, etc.

**Director of Engineering; Xubo Media Ltd; 11/2006 – 12/2010; Hong Kong**

- Provided integration and development services of various systems including CRM, Supply Chain, Product Configuration and Pricing, Content Management, and Call Center;
- Developed and promoted the company's mobile advertising platform ABITO that allowed merchants to post ads to users' handsets based on their thematic subscriptions;
- Developed AdEstate, the system that integrated all advertising available channels and facilitated the trading and management of advertising space, including TV, Radio, Billboard, Bus Wrap, Websites, and Mobile hand set.

**Project Officer; Plateau Homeland Conservation; 9/2005 – 8/2006; Tibet, China**

Coordinated fundraising and sourced resources while managing local participation in Nyingchi to build permanent animal enclosures, improving livestock care beyond seasonal makeshift pens.

**IT Partner; Mercatela Ltd; 2003 - 2005; Tokoyo, Japan**

Developed the market and business for the company with a service model that expanded across Management Consulting, Creative, and IT Implementation; Worked with Sapporo Sparkle and Oracle Japan to build Foodbiz.com for Nikkei Business Publications based on J2EE, JSP, Hibernate, Struts, Oracle DB2, and WebSphere.

**Principal Consultant; IBM Inc; 8/2000 – 7/2003; Burlingame, California**

- Generated requirement documents and proposals in CRM and SCM for Manugistics, BetzDearborn and US West; provided input to project technical staff and transferred knowledge to determine the applicability of IBM CrossWorlds technologies;
- Identified, developed, and analyzed EAI requirements; Developed Technical Requirement Spec (TRS); defined business processes and business objects during the prospect scoping activities across SAP APO, Manugistics BDP, IBM InterChange Server and its infrastructure; defined implementation plans and helped enterprise customers manage risk and maximize returns.

**Sr. Software Engineer; SAP Technology Inc., 1996 – 2000; Palo Alto, California**

Led Sales Force Automation (SFA) design team to deliver function spec and white papers, designed SFA system and the back-end database model; Designed and accomplished RFC Class Library, studied and prototyped CORBA solutions of Iona Orbix and H-ORB, and OLE Controls to heighten RFC availability.

## **SKILLS**

### **Machine Learning & AI**

Agentic AI with Cursor, LangGraph, Claude Code; Deep Learning (CNNs) for image classification and recognitions; Use of LLMs and GPT API to digest, distill, sort, and check similarities; Supervised and Unsupervised methodologies, such as KNN, Random Forest, SVM, Bagging, K-Means, etc; Text Data Processing for text classification, topic modeling,

content analysis.

### **Geospatial Analysis**

Remote Sensing, Radiometric and Geometric Correction, Image Enhancement and Processing, Thematic Information Extraction, Spatial Object Identification and Classification, Change Detection, Internet Mapping and WebGIS, Suitability Modeling, Surface Analysis, Spatial Interpolation, and Areal Interpolation.

### **Programming & Tools**

R, Python, Stata, Php, Java, C++/C#/C; SPSS, Tableau, Power BI; ArcGIS Pro, QGIS, Google Earth Engine, YOLO and RoboFlow for labeling, Drone2Map; database design and data modeling on MySQL, PostgreSQL, and SQLite.

### **Research Design**

Literature Review, Data Collection via surveying and web scrapping, Data Engineering, Statistical Data Modeling and Analysis, Academic Writing.

### **Statistical Analysis**

Statistical and econometric modeling by OLS, GLS, GLM, GLMM, spatial-temporal data modeling by Bayesian/INLA, and Time-Series; Casual inference with DID, SCA, RDD, PSM, and 2SLS; Dimensionality reduction with PCA, Factor Analysis, Ridge and Lasso; Robustness methods by Permutation Testing, Monte Carlo Simulation, and Bootstrapping.

## **EDUCATION**

PhD, Political Science, University of Texas at Dallas, (expected in Fall of 2026)

MS, Geospatial Information Science, University of Texas at Dallas

MS, Social Data Analytics and Research, University of Texas at Dallas

MBA, Strategic Marketing, California State University

MS, Computer Engineering, Ohio State University

MS, Thermodynamics, Shanghai Jiaotong University

BS, Marine Power Engineering, Shanghai Jiaotong University

## **ACHIEVEMENT**

Fred Hill Endowed Scholarship, 2025

Certificate Graduate Teaching, 2024